

YatriRail

INDIAN RAILWAY LIVE INTELLIGENCE AND DYNAMIC ML ETA FORECASTING

Real-time decisions. Accurate arrivals. Smarter journeys.

SUB-5MS

Inference

12-FEATURE

Gradient Boosting
Regressor

8,490

Searchable Indian
Railways Trains

60/40

Operational
Split

DEPLOYMENT STATUS Production-Ready

GITHUB github.com/anuj-maurya-01/YatriRail

LIVE
INTELLIGENCE

DYNAMIC ETA

ML FORECASTING

SITUATIONAL
AWARENESS



THE PROBLEM

Arrival delay uncertainty persists because legacy methods were never built for India's rail network.

Static timetables

Can't adapt to real-time disruptions

Naive delay extrapolation

Assumes delays grow linearly, misses ripple effects

Massive national scale

Thousands of trains, routes, and variables in motion

24M+ commuters impacted

Every day by unpredictable arrivals



What India's rail network needs
A DYNAMIC ML ENGINE
for real-time intelligence and accurate ETAs

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Live Indian Railway Intelligence

01

**Dynamic ML
delay & ETA
forecasting**

Continuously learns.
Predicts in real time.

02

**60/40
operational
split dashboard**

Clear view of running
vs delayed trains.

03

**Network KPIs
and critical alerts**

Track performance.
Act on what matters.

04

**Enterprise
resilience and
zero-downtime
fallback**

Always-on intelligence.
Built for scale.



SYSTEM ARCHITECTURE & END-TO-END DATA FLOW

From live telemetry to **intelligent ETAs** in real time



MACHINE LEARNING ENGINE & FEATURE PIPELINE

12 ENGINEERED FEATURES

Speed
Distance to next station
Historical delay
Time of day
Day of week
Rush hour indicator
Train type
Route congestion
Weather impact
Seasonal pattern
Platform change flag
Signal clearance trend

DYNAMIC ETA FORMULATION

$$\begin{array}{c} \text{ETA} \\ = \\ \text{Timetable arrival} \\ + \\ \text{Live delay} \\ + \\ \text{ML delta} \end{array}$$

MODEL OF CHOICE

HistGradient
Boosting
Regressor

Captures complex
patterns with high
accuracy

Sub-5ms
inference

59
passing tests

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REAL-TIME INTELLIGENCE ENGINE

FRONTEND EXPERIENCE & UI INNOVATION

A 60/40 control-center view
for **real-time operations**

60% Live map

Leaflet-powered network view
with train positions and routes

40% Telemetry panel

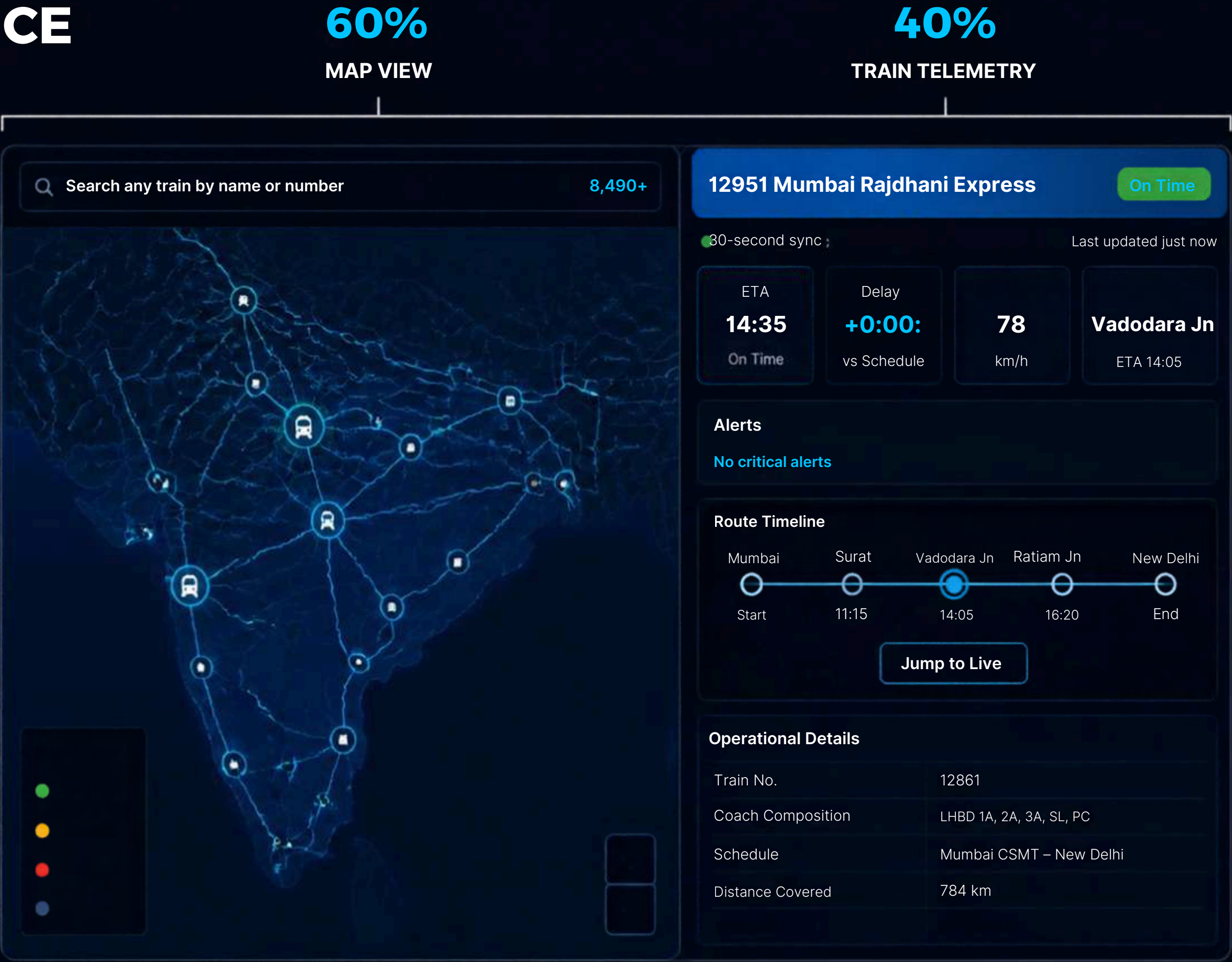
Live KPIs, alerts, and train details
with 30-second sync

Route timeline

Full run view with intermediate stations
and jump to live position

Search across 8,490+ trains

Instantly find any train on the network



Technical Resilience & Edge-Case Engineering

Built to stay accurate, reliable, and secure under real-world constraints.

01

HTTP 429_FALL

Graceful handling when rate limits hit

02

MATHEMATICAL GPS INTERPOLATION

Keeps trajectories smooth when data gaps occur

03

CREDENTIAL ISOLATION

Secure separation of keys and access

04

59 AUTOMATED TESTS

Unit and integration tests that guard every build



KEY IMPACT & HACKATHON ACHIEVEMENTS

8,490+

trains
indexed

<5ms

ML
latency

59/59

tests
passing

100%

cloud
deployed

production
documentation
and [open-source](#)
[MIT repository](#)



FUTURE ROADMAP

Scaling YatriRail Nationally

Three implementation horizons
for expanding the system.

PHASE 01

Build the real-time foundation

- Real-time feeds and weather intelligence
- Mobile PWA for everywhere access
- Multilingual experience for every commuter

PHASE 02

Unlock network intelligence

- Crowd telemetry that learns from every journey
- Platform allocation powered by demand
- Transparent explanations commuters trust

PHASE 03

Scale intelligent operations

- Controller suite for network-wide decisions
- Junction precedence for smoother flow
- Energy optimization and fleet analytics



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Predictable railway travel

THANK YOU

Questions & Answers

A dedicated team building live intelligence for India's rail network

Aru Shubham Singh

Machine Learning
Engineer

Archita Kesharwani

Backend &
Data Engineer

Anuj Maurya

Frontend &
Product Engineer

Shlok Bhardwaj

Data Analytics &
Research

Public Repository

google.com/team/yatirail

Deployed On

Render & Vercel

Test Coverage

59 automated
tests passing

